



University College Roosevelt

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Bachelor of Liberal Arts & Sciences

Senior Project

Photon Orbits and the Schwarzschild Black Hole Shadow

A numerical ray-tracing study of the capture–escape boundary for null geodesics

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Abstract

This thesis investigates how null geodesics in Schwarzschild spacetime produce the capture–escape boundary that defines an idealised black-hole shadow. The motivation comes from black-hole imaging, where the shadow is not a direct image of the event horizon, but the set of observer directions whose corresponding light rays are captured rather than connected to external sources.

The project studies this boundary by deriving the Schwarzschild null-geodesic equations and implementing a Python ray tracer for photon trajectories near the known critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$. The full geodesic-equation formulation is first used to show the connection between the metric and the photon trajectory. A reduced formulation based on conserved quantities is then used for the numerical implementation, because it makes the impact parameter, radial turning points, and branch switching more direct.

The simulation reproduces the expected separation between captured and escaping rays. Near the boundary, photons pass close to the unstable photon sphere and accumulate large angular displacement. This behaviour appears in the trajectory plots, winding estimates, and finite-radius corrected deflection angles. The results show that the Schwarzschild shadow boundary is set by photon capture, not by the horizon radius itself.

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1 Introduction

1.1 Motivation and Scientific Context

General Relativity describes gravity as spacetime curvature produced by mass and energy. In this theory, a black hole is bounded by an event horizon, beyond which no future-directed signal can return to distant observers. Light is therefore one of the main ways to study the exterior geometry. In the geometric-optics limit, photons follow null geodesics, so their paths are determined by the spacetime metric [1, 2].

My project began with Project B in Taylor, Wheeler, and Bertschinger's *Exploring Black Holes*, which asks what the sky would look like to an observer falling into a Schwarzschild black hole [2]. I first approached this as a visual question: what does a black hole look like? That question quickly became a question about photon paths. Since light from inside the event horizon cannot reach distant observers, the apparent darkness of a black hole has to be understood through the rays that fail to connect to external light sources.

This shift made the problem computational. A black-hole image is not built from one photon path. Each point on an observer's screen corresponds to a different possible null geodesic. Some of these geodesics escape to external regions, while others are captured by the black hole. The visual boundary of the shadow is therefore linked to a transition between two possible photon outcomes: escape and capture.

Modern black-hole imaging gives this geometric question direct observational relevance. The Event Horizon Telescope Collaboration reconstructed ring-like emission structures with central brightness depressions around M87* and Sagittarius A* [3, 4]. These images are not simple photographs of event horizons. They are shaped by emission near the compact object, gravitational lensing, photon capture, and the finite resolution of the telescope array. For that reason, the geodesic structure of the spacetime is a necessary starting point for understanding why black-hole images can contain dark central regions and bright surrounding emission [5, 6].

The Schwarzschild black hole is a useful starting point because it removes many complications. It is not a complete model of an astrophysical black-hole image. It excludes rotation, accretion physics, plasma effects, and telescope reconstruction. Its value is that it isolates the geometric mechanism behind photon capture in the simplest black-hole spacetime. The same model still contains gravitational deflection, unstable photon motion, and a well-defined circular shadow boundary [7, 8, 6].

The relevance of this project is therefore not that it produces a realistic image of M87* or Sagittarius A*. It is that numerical ray tracing makes the capture–escape structure visible and testable. By following photon paths directly, the shadow can be treated as a concrete computational boundary: which rays escape, and which rays are captured?

1.2 From Photon Orbits to a Numerical Problem

The motivation above leads to a more precise problem. If each screen direction corresponds to a possible photon path, then the shadow can be studied by tracing those paths and checking their final outcomes. This turns the visual question into a ray-classification problem.

Schwarzschild spacetime is well suited to this calculation. Its spherical symmetry allows photon motion to

be restricted to a plane, and each trajectory can be labelled by an impact parameter b . This parameter controls the angular momentum of the photon relative to its energy. In the numerical ray tracer, b becomes the main input variable.

The analytic theory predicts a critical impact parameter,

$$b_{\text{crit}} = 3\sqrt{3}M.$$

This value separates the two photon outcomes for rays arriving from far away. Rays below the critical value are captured, while rays above it reach a closest approach and escape. Near this boundary, photons can spend a long angular interval close to the unstable photon sphere before their final outcome becomes clear [7, 8, 9].

This near-critical behaviour is the part of the problem that motivated the simulation. The analytic value of b_{crit} is known, but a ray tracer still has to reproduce the transition correctly. The code must handle inward and outward radial branches, radial turning points, and numerical sensitivity near the boundary. These are practical issues that appear when the geodesic equations are turned into an actual computation.

The connection to the black-hole shadow then follows from the classification. In backward ray tracing, a screen direction belongs to the idealised shadow if its corresponding photon trajectory is captured. If the trajectory escapes to an external source region, that screen direction can receive light [10, 6]. In this sense, the shadow boundary is the screen-space version of the capture–escape boundary.

This sets up the numerical task of the thesis. I trace Schwarzschild null geodesics for different values of b , classify each trajectory as captured or escaping, and examine how the near-critical paths behave close to b_{crit} . The same classification is then used to construct the idealised Schwarzschild shadow.

1.3 Aim and Research Question

The aim of this thesis is to investigate the capture–escape boundary for null geodesics in Schwarzschild spacetime using numerical ray tracing. I focus on trajectories close to the critical impact parameter, where small changes in b can lead to different final outcomes.

The central research question is:

How can numerical ray tracing of near-critical null geodesics in Schwarzschild spacetime reproduce the transition between photon capture and escape, and how does this transition define the boundary of an idealised black-hole shadow?

I used the analytic value $b_{\text{crit}} = 3\sqrt{3}M$ as a benchmark, not as an unknown to be discovered numerically. The task was to test whether the ray tracer reproduced the expected capture–escape split and to study how photon trajectories behave near that boundary.

To do this, I derived the Schwarzschild null-geodesic equations in two forms. The full geodesic-equation formulation gives the direct link between the metric and the photon trajectory. The reduced conserved-quantity formulation expresses the same motion in terms of the impact parameter and radial turning points. I used the reduced formulation in the numerical code because it made the capture–escape classification more direct.

The numerical results follow the research question directly. The trajectory plots show the separation between captured and escaping rays. The angular displacement and winding estimates show the near-critical behaviour close to the photon sphere. The deflection-angle calculation connects the simulation to weak-field and strong-field lensing. The shadow construction then translates the capture–escape classification into an idealised Schwarzschild shadow.

The thesis is organised around this calculation. Section 2 derives the theoretical framework needed for the ray tracer. Section 3 presents the numerical implementation and results together, including the algorithm, branch switching, trajectory classifications, winding estimates, deflection angles, and shadow construction. Section 4 interprets the results and discusses the scope of the model. Section 5 summarises the findings and gives possible extensions.

The project therefore starts from a visual question, but reduces it to a computational one: trace the null geodesics, identify which ones are captured, and use that boundary to understand the idealised Schwarzschild shadow.

2 Theoretical Background

This section develops the theoretical framework used in the numerical ray tracing. A focus on the ingredients needed for the computation: null geodesics, the impact parameter, radial turning points, the photon sphere, the critical impact parameter, gravitational deflection, and the idealised Schwarzschild shadow [1, 2, 6].

The calculation starts from Schwarzschild spacetime, which describes the exterior vacuum geometry of a static, spherically symmetric, non-rotating black hole [1, 2]. In the geometric-optics approximation, photons follow null geodesics. I used two equivalent formulations of these geodesics. The first uses the full geodesic equation and the Christoffel symbols of the Schwarzschild metric. This formulation gives the most direct link between the metric and the photon trajectory. The second uses the conserved energy and angular momentum of Schwarzschild null geodesics. This reduces the equations to a form in which the impact parameter $b = L/E$ appears explicitly.

Both formulations describe the same photon paths. They are included for different reasons. The full geodesic equation shows the geometric origin of the motion and was the natural starting point for the project. The reduced conserved-quantity formulation is better suited to the numerical implementation, because the research question depends on varying the impact parameter and classifying trajectories as captured or escaping. It also makes radial turning points explicit, which is useful when integrating near-critical rays.

The section therefore builds toward the reduced equations used in Section 3. After deriving the equations of motion, I use the effective-potential picture to obtain the photon sphere at $r = 3M$ and the critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$. These results set the analytic benchmark for the numerical ray tracing. The final part of the section connects this capture–escape boundary to gravitational deflection and to the idealised Schwarzschild black-hole shadow.

2.1 Schwarzschild Spacetime

The spacetime considered in this thesis is Schwarzschild spacetime. It is the idealised solution for a static, spherically symmetric, uncharged, and non-rotating black hole [1, 2]. This model isolates the geometric effects relevant to the project: photon capture, gravitational deflection, unstable photon orbits, and a circular shadow boundary.

I used geometric units throughout the thesis, so that

$$G = c = 1.$$

This choice removes repeated factors of G and c from the equations. In these units, mass can be measured as a length. The quantity GM/c^2 , which has dimensions of length in SI units, is written simply as M . All radii and impact parameters in the calculation are therefore expressed in units of the black-hole mass.

The Schwarzschild metric is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (1)$$

This line element tells us how coordinate differences combine into the spacetime interval ds^2 . The factor $1 - 2M/r$ carries the gravitational effect of the black hole in these coordinates. Here t is the Schwarzschild time coordinate, r is the Schwarzschild radial coordinate, and θ and ϕ are angular coordinates. The coordinate r is an areal radius: a sphere with constant r has surface area $4\pi r^2$. This is why radii such as the event horizon and photon sphere are quoted using the Schwarzschild radial coordinate.

The event horizon is located at

$$r = 2M. \quad (2)$$

At this radius, the factor $1 - 2M/r$ vanishes. In Schwarzschild coordinates, the coefficient of dt^2 becomes zero and the coefficient of dr^2 diverges. This divergence is a coordinate singularity, not the curvature singularity of the spacetime. The physical singularity lies at $r = 0$ [1, 2]. In the numerical work, captured rays were therefore stopped just outside $r = 2M$, where the trajectory outcome is already clear for the exterior problem.

Two symmetries of the Schwarzschild metric are used throughout the thesis. First, the metric does not depend explicitly on t , so the spacetime is stationary. This symmetry gives a conserved photon energy. Second, the metric is spherically symmetric. This symmetry gives a conserved angular momentum and allows each photon trajectory to be placed in a plane through the centre of the black hole.

I chose that plane to be the equatorial plane,

$$\theta = \frac{\pi}{2}.$$

This is a coordinate choice, not an additional physical assumption. With $\theta = \pi/2$ and $d\theta = 0$, Eq. (1) reduces to

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\phi^2. \quad (3)$$

This reduced metric contains the time, radial, and angular parts needed to describe photon motion in the orbital plane. It is the starting point for the null-geodesic equations derived next.

2.2 Null Geodesics and the Affine Parameter

In General Relativity, freely moving particles and light rays follow geodesics of the spacetime metric. Massive particles follow timelike geodesics, while photons follow null geodesics. In the geometric-optics approximation, a photon can therefore be treated as a light ray whose trajectory satisfies the null condition $ds^2 = 0$ [1, 2]. This approximation turns the problem of light propagation into a problem of solving geodesic equations.

A photon trajectory is written as a curve through spacetime,

$$x^\mu(\lambda) = (t(\lambda), r(\lambda), \theta(\lambda), \phi(\lambda)),$$

where λ labels points along the path. The tangent vector to this curve is

$$k^\mu = \frac{dx^\mu}{d\lambda}. \quad (4)$$

The vector k^μ gives the rate at which the photon moves through each coordinate as the parameter λ changes. For massive particles, proper time can be used as the natural parameter along the path. For photons, the proper time along the trajectory is zero, so a different parameter is needed. An affine parameter λ is introduced for this purpose.

The affine parameter is not the time measured by a physical observer. It is a mathematical parameter used to write the photon trajectory as a set of differential equations. Its absolute scale is not physically important here. A constant rescaling of λ describes the same geometric path with a different parametrisation. This freedom is used later when the photon energy scale is set to $E = 1$ in the reduced equations.

The null condition follows from the spacetime interval,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (5)$$

Dividing by $d\lambda^2$ and using $k^\mu = dx^\mu/d\lambda$ gives

$$\left(\frac{ds}{d\lambda}\right)^2 = g_{\mu\nu} k^\mu k^\nu.$$

For a photon, $ds^2 = 0$, so the tangent vector must satisfy

$$g_{\mu\nu} k^\mu k^\nu = 0. \quad (6)$$

This equation is the null constraint on the photon trajectory [1]. It states that the photon tangent vector has zero spacetime length, which is the geometric meaning of lightlike motion.

Using the equatorial Schwarzschild metric from Eq. (3), the null condition becomes

$$0 = -\left(1 - \frac{2M}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2. \quad (7)$$

This constraint links the time, radial, and angular components of the photon motion. They cannot be chosen independently if the trajectory is to remain null. For example, a larger angular component changes the radial motion allowed by the same null condition.

Equation (7) is not yet an equation of motion. It only restricts which tangent vectors are allowed for a photon. To determine how the trajectory evolves, the geodesic equation must also be imposed. The next subsection introduces this full geodesic-equation formulation, which gives the most direct route from the Schwarzschild metric to the photon equations of motion.

2.3 Formulation I: The Full Geodesic Equation

The first formulation uses the full geodesic equation. This is the most direct way to compute photon motion from the metric, because the curvature information enters through the Christoffel symbols. Once the Schwarzschild metric is known, the geodesic equation determines how a freely moving photon trajectory evolves through that geometry [1, 2].

For an affine parameter λ , the geodesic equation is

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0. \quad (8)$$

The first term describes the coordinate acceleration of the curve. The second term corrects that acceleration for the curvature and coordinate structure of spacetime. A geodesic is therefore the curve whose apparent coordinate acceleration is exactly balanced by the connection terms.

Here $x^\mu(\lambda)$ is the photon trajectory and $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols. These are computed from the metric by

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta}). \quad (9)$$

The Christoffel symbols contain first derivatives of the metric. They encode how the coordinate basis changes from point to point, and they are the terms through which the Schwarzschild geometry enters the equations of motion.

Since the motion has been restricted to the equatorial plane, the relevant coordinates are t , r , and ϕ . The photon trajectory can therefore be written as

$$x^\mu(\lambda) = (t(\lambda), r(\lambda), \phi(\lambda)).$$

I introduced the tangent-vector components

$$k^t = \frac{dt}{d\lambda}, \quad k^r = \frac{dr}{d\lambda}, \quad k^\phi = \frac{d\phi}{d\lambda}. \quad (10)$$

This converts the second-order geodesic equation into a first-order system for the coordinates and tangent-vector components. That form is closer to what a numerical ODE solver expects.

For the equatorial Schwarzschild metric in Eq. (3), the required non-zero Christoffel symbols are

$$\Gamma_{tr}^t = \Gamma_{rt}^t = \frac{M}{r(r-2M)}, \quad (11)$$

$$\Gamma_{tt}^r = \frac{M(r-2M)}{r^3}, \quad (12)$$

$$\Gamma_{rr}^r = -\frac{M}{r(r-2M)}, \quad (13)$$

$$\Gamma_{\phi\phi}^r = -(r-2M), \quad (14)$$

$$\Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = \frac{1}{r}. \quad (15)$$

These terms determine how the time, radial, and angular components of the photon tangent vector exchange with each other as the photon moves through the Schwarzschild geometry.

Substitution into Eq. (8) gives the coordinate equations

$$\frac{dt}{d\lambda} = k^t, \quad (16)$$

$$\frac{dr}{d\lambda} = k^r, \quad (17)$$

$$\frac{d\phi}{d\lambda} = k^\phi. \quad (18)$$

These equations define the tangent-vector components as the derivatives of the coordinates. The remaining equations describe how those components evolve.

The tangent-vector components evolve according to

$$\frac{dk^t}{d\lambda} = -2 \frac{M}{r(r-2M)} k^t k^r, \quad (19)$$

$$\frac{dk^r}{d\lambda} = -\frac{M(r-2M)}{r^3} (k^t)^2 + \frac{M}{r(r-2M)} (k^r)^2 + (r-2M)(k^\phi)^2, \quad (20)$$

$$\frac{dk^\phi}{d\lambda} = -\frac{2}{r} k^r k^\phi. \quad (21)$$

The radial equation contains the main competition in the problem. The terms involving k^t , k^r , and k^ϕ determine whether the photon continues inward, turns around, or approaches the near-critical region. Equations (16)–(21) are the full geodesic-equation formulation for equatorial photon motion in Schwarzschild spacetime.

The tangent vector must also satisfy the null constraint. Using Eq. (3), this constraint is

$$0 = -\left(1 - \frac{2M}{r}\right) (k^t)^2 + \left(1 - \frac{2M}{r}\right)^{-1} (k^r)^2 + r^2 (k^\phi)^2. \quad (22)$$

This is not an additional evolution equation. It is a constraint on the initial tangent vector, and it can also be used as a numerical check on the integration. If the integration is accurate, the left-hand side should remain close to zero along the trajectory.

This formulation was the natural starting point for the project because it follows directly from the metric. It is also the more general route: the same method can be applied to other spacetimes once the metric and Christoffel symbols are known. For the specific Schwarzschild ray-tracing problem in this thesis, however, the full formulation evolves more variables than are needed for the spatial trajectory. It also hides the impact parameter, which is the main control variable in the research question. The next subsection therefore uses the conserved quantities of Schwarzschild spacetime to write the same photon motion in a form better suited to capture–escape classification.

2.4 Formulation II: Conserved Quantities and the Reduced Equations

The Schwarzschild metric has symmetries that simplify the photon equations of motion. Since the metric does not depend explicitly on t , there is a conserved energy. Since it does not depend explicitly on ϕ , there is a conserved angular momentum. For equatorial null geodesics, these quantities are [1, 2]

$$E = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\lambda}, \quad (23)$$

and

$$L = r^2 \frac{d\phi}{d\lambda}. \quad (24)$$

The conserved energy E controls the time component of the photon motion, while L controls the angular motion around the black hole. Because both quantities remain constant along a geodesic, they allow the photon trajectory to be described without evolving the full set of coordinate and tangent-vector equations from Formulation I.

The ratio of angular momentum to energy is the impact parameter,

$$b = \frac{L}{E}. \quad (25)$$

For a photon arriving from far from the black hole, b corresponds to the perpendicular distance between the incoming asymptotic path and the centre of the black hole. In this thesis, b is the main control parameter. Varying b changes whether a photon is captured by the black hole or escapes after reaching a turning point.

From Eqs. (23) and (24),

$$\frac{dt}{d\lambda} = \frac{E}{1 - \frac{2M}{r}}, \quad \frac{d\phi}{d\lambda} = \frac{L}{r^2}. \quad (26)$$

These relations replace the time and angular equations with algebraic expressions involving the constants E and L . Substituting them into the equatorial null condition in Eq. (7) gives

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{2M}{r}\right) \frac{L^2}{r^2}. \quad (27)$$

This equation is the radial motion equation. It shows that the radial part of the photon motion is set by a balance between the energy term E^2 and the angular barrier term proportional to L^2/r^2 .

Using $b = L/E$, this becomes

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 \left[1 - \left(1 - \frac{2M}{r}\right) \frac{b^2}{r^2}\right]. \quad (28)$$

Now the impact parameter appears explicitly. This is the form that makes a scan over different photon trajectories natural.

The overall scale of the affine parameter is arbitrary. Rescaling λ changes the numerical value of E , but not the spatial path of the photon. I therefore set

$$E = 1$$

in the numerical work. With this choice, $L = b$, and Eq. (28) reduces to

$$\left(\frac{dr}{d\lambda}\right)^2 = 1 - \left(1 - \frac{2M}{r}\right) \frac{b^2}{r^2}. \quad (29)$$

This normalisation fixes the parametrisation of the ray but leaves the spatial trajectory unchanged. It also makes b the only trajectory parameter that I vary in the ray-tracing calculation.

For the numerical implementation, it is useful to name the right-hand side of Eq. (29). I define

$$\mathcal{R}(r; b) = 1 - \left(1 - \frac{2M}{r}\right) \frac{b^2}{r^2}. \quad (30)$$

This quantity is the radial radicand, since it appears under the square root when solving for $dr/d\lambda$. It matches the role of the `radial_radicand` function in the Python implementation.

Real radial motion requires

$$\mathcal{R}(r; b) \geq 0. \quad (31)$$

This inequality marks the radii that the photon can physically access for a chosen value of b . A radial turning point occurs when

$$\mathcal{R}(r; b) = 0. \quad (32)$$

At this point $dr/d\lambda = 0$. For an escaping photon, this is the radius of closest approach. The photon first moves inward, reaches the turning point, and then moves outward again.

The final reduced equations for the spatial trajectory are

$$\frac{dr}{d\lambda} = s_r \sqrt{\mathcal{R}(r; b)}, \quad (33)$$

$$\frac{d\phi}{d\lambda} = \frac{b}{r^2}. \quad (34)$$

The sign s_r specifies the radial branch. For inward motion, $s_r = -1$. For outward motion, $s_r = +1$. This sign is needed because Eq. (29) determines only $(dr/d\lambda)^2$, not the sign of $dr/d\lambda$. When an escaping photon reaches a turning point, the integration must switch from the inward branch to the outward branch.

Equations (33) and (34), together with Eq. (30), are the reduced equations used in the final ray-tracing code. The Schwarzschild coordinate time can be recovered from Eq. (26), but it is not needed for the trajectory plots and capture–escape classification used in this thesis.

2.4.1 Choice of the Reduced Formulation

The full and reduced formulations describe the same null geodesics. The difference is practical rather than physical. Formulation I follows directly from the geodesic equation and shows explicitly how the metric produces photon motion. It was therefore the appropriate starting point for the project. It also remains the more general method, since it can be applied to other spacetimes once the metric is known.

For the final numerical ray tracing, I used the reduced conserved-quantity formulation. The reason is that the research question is organised around the impact parameter. In Formulation II, $b = L/E$ appears directly in the equations. This makes it straightforward to scan over different impact parameters, compare trajectories just below and above b_{crit} , and classify each ray as captured or escaping.

The reduced formulation also gives a clear numerical condition for radial turning points:

$$\mathcal{R}(r; b) = 0.$$

This matters because escaping trajectories do not move inward forever. They approach the black hole, reach a closest radius, and then return outward. In the code, this behaviour is handled by integrating the inward and outward parts as separate branches, with $s_r = -1$ before the turning point and $s_r = +1$ after it.

This branch treatment is especially important near the critical impact parameter. Near-critical photons can remain close to the unstable photon sphere for a long angular interval. In that regime, small numerical errors near a turning point can noticeably affect the computed angular displacement. The reduced formulation makes the sensitive quantities explicit: the impact parameter b , the radial radicand $\mathcal{R}(r; b)$, and the branch sign s_r . For that reason, it is the formulation used in the numerical ray-tracing results in Section 3.

2.5 Effective Potential, Photon Sphere, and Critical Impact Parameter

The reduced radial equation can be written in an energy-balance form. Starting from Eq. (27),

$$\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}^2(r) = E^2, \quad (35)$$

where

$$V_{\text{eff}}^2(r) = \left(1 - \frac{2M}{r}\right) \frac{L^2}{r^2}. \quad (36)$$

I write this term as V_{eff}^2 because it appears in the same equation as E^2 . The equation has the form of an energy balance: the radial kinetic term plus an effective potential term equals the conserved energy squared. This form shows which radii are accessible to a photon with fixed energy E and angular momentum L . Radial motion is possible only where

$$E^2 \geq V_{\text{eff}}^2(r).$$

The same effective-potential picture is commonly used to discuss photon orbits in Schwarzschild space-time [1, 2].

A radial turning point occurs when the radial velocity vanishes. From Eq. (35), this condition is

$$E^2 = V_{\text{eff}}^2(r). \quad (37)$$

At such a radius, all of the available energy is in the effective-potential term, so the radial motion momentarily stops. In the reduced formulation, the same condition is

$$\mathcal{R}(r; b) = 0, \quad (38)$$

because $\mathcal{R}(r; b)$ is the radicand in Eq. (33). Substituting Eq. (30) gives

$$1 - \left(1 - \frac{2M}{r}\right) \frac{b^2}{r^2} = 0. \quad (39)$$

Rearranging this expression gives the cubic equation

$$r^3 - b^2 r + 2Mb^2 = 0. \quad (40)$$

For an escaping trajectory, the largest real root outside the event horizon gives the radius of closest approach. This is the turning radius used later when the numerical integration switches from the inward to the outward radial branch.

A circular photon orbit requires more than $dr/d\lambda = 0$ at one point. The radius must also remain fixed under the radial evolution. In the effective-potential picture, this means that $V_{\text{eff}}^2(r)$ must have an extremum:

$$\frac{d}{dr} V_{\text{eff}}^2(r) = 0. \quad (41)$$

This condition identifies radii where a photon can remain at constant r , rather than merely pausing at a turning point.

Using Eq. (36),

$$\frac{d}{dr} \left[\left(1 - \frac{2M}{r}\right) \frac{L^2}{r^2} \right] = \frac{2L^2}{r^4} (3M - r). \quad (42)$$

The factor $3M - r$ shows that there is one circular photon radius outside the horizon. The derivative vanishes at

$$r = 3M. \quad (43)$$

This radius is the Schwarzschild photon sphere. It lies outside the event horizon at $r = 2M$ and corresponds to unstable circular photon motion. The instability follows from the fact that $V_{\text{eff}}^2(r)$ has a maximum at $r = 3M$. A photon exactly on this orbit remains there only in the limiting mathematical case. A small perturbation sends it either inward toward capture or outward toward escape [7, 9, 6].

The critical impact parameter follows by evaluating the turning-point condition at the photon sphere. At the critical circular photon orbit,

$$r = 3M, \quad E^2 = V_{\text{eff}}^2(3M).$$

Since

$$1 - \frac{2M}{3M} = \frac{1}{3},$$

Eq. (36) gives

$$E^2 = \frac{L^2}{27M^2}. \quad (44)$$

This relation is the threshold between photons that can turn around and photons that are captured. Using $b = L/E$, the critical impact parameter is therefore

$$b_{\text{crit}} = 3\sqrt{3}M. \quad (45)$$

For photons arriving from far away, this value separates the two possible outcomes. Photons with $b < b_{\text{crit}}$ are captured by the black hole. Photons with $b > b_{\text{crit}}$ reach an outer turning point and escape after being deflected. At $b = b_{\text{crit}}$, the limiting trajectory asymptotically approaches the unstable photon sphere [7, 6].

The critical impact parameter should not be confused with a radius in space. The photon sphere is located at the Schwarzschild coordinate radius $r = 3M$. The critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$ labels photon trajectories as seen from large distance or on an observer's image screen. It is the threshold value of b , not a material surface around the black hole. This distinction matters because the Schwarzschild shadow boundary is set by b_{crit} , not by the horizon radius $2M$.

2.6 Gravitational Deflection and Near-Critical Behaviour

Gravitational deflection is one of the standard observational consequences of General Relativity. In Schwarzschild spacetime, the bending depends mainly on the impact parameter b . A photon with large b passes far from the black hole and is only weakly deflected. A photon with b close to b_{crit} passes close to the photon sphere and can undergo much stronger bending [1, 2, 7].

The deflection angle is defined for escaping photons. These are trajectories with $b > b_{\text{crit}}$. They move inward from large radius, reach a closest approach, and then return outward. In the reduced formulation, the closest approach is a radial turning point. Since

$$\frac{dr}{d\lambda} = s_r \sqrt{\mathcal{R}(r; b)},$$

the turning point r_{min} satisfies

$$\mathcal{R}(r_{\text{min}}; b) = 0. \quad (46)$$

This condition is the same turning-point condition used later in the numerical branch switch. It identifies the radius where the photon changes from inward to outward radial motion.

The total angular change of an escaping trajectory can be written using the reduced equations. Dividing $d\phi/d\lambda$ by $dr/d\lambda$ gives

$$\left| \frac{d\phi}{dr} \right| = \frac{b}{r^2 \sqrt{\mathcal{R}(r; b)}}. \quad (47)$$

This expression gives the angular change per unit radial change. It becomes large when $\mathcal{R}(r; b)$ becomes small, which is why rays near a turning point can accumulate significant angular displacement. The absolute value is used because the total angular change counts both the inward and outward parts of the trajectory.

For a photon that starts at infinity, reaches r_{min} , and escapes back to infinity, the total angular displacement is

$$\Delta\phi = 2 \int_{r_{\text{min}}}^{\infty} \frac{b}{r^2 \sqrt{\mathcal{R}(r; b)}} dr. \quad (48)$$

The factor of two accounts for the two radial branches of the scattering path. This integral increases sharply when the photon passes close to the photon sphere, because the denominator becomes small over a longer part of the path.

In flat spacetime, the corresponding angular change for a straight-line scattering trajectory is π . The Schwarzschild deflection angle is therefore

$$\delta\phi = \Delta\phi - \pi. \quad (49)$$

This is the asymptotic definition of the deflection angle. It assumes that the photon begins and ends infinitely far from the black hole [8, 6]. In the numerical implementation, the start and escape radii are finite, so I subtract the angular change of a flat-space trajectory over the same finite radial interval. This gives the finite-radius corrected deflection angle used in Section 3.

In the weak-field limit, $M/b \ll 1$, the Schwarzschild deflection angle can be expanded in powers of M/b . The leading-order result is

$$\delta\phi \approx \frac{4M}{b}. \quad (50)$$

This expression shows that the bending decreases as the impact parameter increases. It describes photons that remain far from the black hole, where the gravitational field is weak [1, 2]. A more accurate weak-field expression includes the second-order term,

$$\delta\phi \approx \frac{4M}{b} + \frac{15\pi M^2}{4b^2}. \quad (51)$$

The second term gives the next correction in the weak-field expansion. It improves the comparison at moderately large b , but it should not be expected to describe rays near the photon sphere [11, 8].

The weak-field expansion breaks down as b approaches b_{crit} from above. In this regime, the denominator in Eq. (47) becomes small near the closest approach, and the photon can spend a long angular interval close to $r = 3M$. Darwin studied this strong bending for Schwarzschild trajectories, and Bozza later developed a general strong-field lensing expansion in which the deflection angle diverges logarithmically near the capture threshold [7, 8].

For $b < b_{\text{crit}}$, the photon is captured rather than scattered back to large radius. In that case, the asymptotic deflection angle in Eq. (49) is not defined. The photon never reaches an outgoing asymptotic branch. The total angular change $\Delta\phi$ can still be measured along the computed path, however. I used that quantity later to compare near-critical captured and escaping rays.

The numerical results use these ideas in two ways. For escaping photons, I compute a finite-radius corrected deflection angle and compare it with Eqs. (50) and (51). For both captured and escaping near-critical photons, I compute the total angular displacement and the approximate winding number. These quantities show how the motion changes from ordinary weak bending to near-critical winding around the black hole.

2.7 The Schwarzschild Black-Hole Shadow

The previous subsections showed that Schwarzschild null geodesics are separated by the critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$. This same capture–escape boundary defines the idealised Schwarzschild

black-hole shadow. In this thesis, the shadow is the set of directions on an observer's screen whose corresponding null geodesics are captured by the black hole. Equivalently, in a backward ray-tracing picture, a screen direction belongs to the shadow if the ray traced backward from the observer terminates at the black hole rather than reaching an external light source [10, 6].

For a Schwarzschild black hole observed from large distance, the shadow is circular. This follows from spherical symmetry: no direction around the black hole is preferred [6]. In the impact-parameter screen used in the numerical construction, I write the horizontal and vertical screen coordinates as b_x and b_y . The distance from the centre of the screen is then

$$B = \sqrt{b_x^2 + b_y^2}. \quad (52)$$

The quantity B is not a Schwarzschild radial coordinate. It is a radius in observer-screen space, measured in units of impact parameter. It tells us how far a screen point lies from the centre of the apparent image.

In the Schwarzschild case, the capture condition on this screen is

$$B < b_{\text{crit}}. \quad (53)$$

These screen directions correspond to photon trajectories that are captured by the black hole. Directions with $B > b_{\text{crit}}$ correspond to escaping trajectories. The boundary of the idealised Schwarzschild shadow is therefore

$$B = b_{\text{crit}} = 3\sqrt{3}M. \quad (54)$$

This is the theoretical basis for the shadow grid constructed in the numerical section: pixels inside $B = b_{\text{crit}}$ are assigned to the captured region, while pixels outside this boundary are assigned to escaping directions.

This result also shows why the shadow is not a direct image of the event horizon. The event horizon is the causal boundary at $r = 2M$. The photon sphere is the unstable circular photon orbit at $r = 3M$. The critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$ is the capture–escape threshold for null geodesics. The shadow is the corresponding dark region on the observer's screen. These four objects are related, but they are not the same object [9, 6]. This distinction is important because the trajectory plots in this thesis are drawn in Schwarzschild coordinate space, while the shadow plot is drawn in impact-parameter screen space.

For an observer at a finite Schwarzschild radius, the shadow can also be described by an angular radius. Let r_{obs} be the radial position of a static observer outside the photon sphere. The angular radius α_{sh} is the angle between the inward radial direction and the direction of the shadow edge on the observer's sky. For Schwarzschild spacetime, analytical treatments of black-hole shadows give

$$\sin \alpha_{\text{sh}} = \frac{b_{\text{crit}}}{r_{\text{obs}}} \sqrt{1 - \frac{2M}{r_{\text{obs}}}}. \quad (55)$$

This expression relates the same critical impact parameter to the angular size seen by a local static observer [6]. The factor $b_{\text{crit}}/r_{\text{obs}}$ gives the geometric scaling with distance, while the square-root factor accounts for the Schwarzschild geometry at the observer position. At large observer radius, the small-angle limit gives

$$\alpha_{\text{sh}} \approx \frac{b_{\text{crit}}}{r_{\text{obs}}}. \quad (56)$$

This limit shows the expected inverse-distance behaviour: a more distant observer sees a smaller angular shadow. I use Eq. (55) later to plot how the apparent angular radius of the Schwarzschild shadow changes with observer distance.

The shadow boundary is closely connected to strong gravitational lensing. Rays near $B = b_{\text{crit}}$ have impact parameters close to the capture threshold, so they pass near the photon sphere and can accumulate large angular displacement before escaping. In images with emitting matter, such near-critical trajectories are related to higher-order lensing features and photon-ring-like structures [9, 6]. The present calculation focuses on the geodesic part of this structure: which photon directions are captured, which escape, and where the boundary lies.

The link between photon motion and black-hole appearance has been studied in more realistic visual models. Luminet showed how strong lensing changes the apparent image of a spherical black hole surrounded by a thin accretion disc [5]. Modern shadow reviews describe how observer position, photon regions, light rings, and spacetime geometry control the observed shadow boundary [9, 6]. The Schwarzschild calculation used here is the geometric core of that broader problem.

This completes the theoretical chain used in the numerical work. The Schwarzschild metric gives the null-geodesic equations. The conserved quantities reduce those equations to a two-dimensional ray-tracing system. The effective-potential analysis gives the photon sphere and the critical impact parameter. Finally, the critical impact parameter gives the capture–escape boundary that defines the idealised shadow. With these pieces established, the next step is to implement the reduced equations numerically, trace rays for different values of b , and compare the computed trajectories with the analytic boundary.

3 Numerical Ray Tracing and Results

Section 2 established that the Schwarzschild shadow can be treated as a capture–escape problem for null geodesics. The critical impact parameter $b_{\text{crit}} = 3\sqrt{3}M$ gives the analytic boundary, while the reduced equations describe how each photon trajectory evolves for a chosen value of b . The numerical task was therefore direct: I integrated the reduced equations, classified each ray, and then used the resulting trajectories to interpret the shadow boundary.

This section presents the implementation and the results together. That structure is useful because each numerical output follows from a specific part of the algorithm. The branch-switching procedure determines how escaping rays are integrated through a turning point. The capture–escape classification tests the analytic value of b_{crit} . The near-critical trajectories and deflection angles then show why the shadow boundary is physically meaningful, rather than just a line drawn in impact-parameter space.

The calculation uses the reduced equations

$$\begin{aligned}\frac{dr}{d\lambda} &= s_r \sqrt{\mathcal{R}(r; b)}, \\ \frac{d\phi}{d\lambda} &= \frac{b}{r^2},\end{aligned}$$

with the radial radicand $\mathcal{R}(r; b)$ defined in Eq. (30). These equations are the computational form of the theory developed in Section 2. The impact parameter b is the main input variable, while the sign s_r tracks whether the photon is on the inward or outward radial branch.

The analytic value

$$b_{\text{crit}} = 3\sqrt{3}M$$

is used as a benchmark throughout. I did not use the code to discover this value independently. Instead, I used it to test whether the ray tracer reproduces the expected behaviour on either side of the capture–escape boundary. This makes the numerical results a validation and interpretation of the theory from Section 2.

3.1 Numerical Set-up and Ray-Tracing Algorithm

The ray-tracing algorithm has three roles. First, it turns the reduced geodesic equations into a first-order system that can be integrated numerically. Second, it handles the change of radial direction for escaping photons. Third, it assigns each trajectory to a physical outcome: capture or escape. These three steps are connected. Without the branch switch at the turning point, the code cannot reliably follow escaping rays; without reliable escaping rays, the capture–escape boundary cannot be tested.

3.1.1 Initial conditions and numerical parameters

The numerical code evolves the photon coordinates $r(\lambda)$ and $\phi(\lambda)$ as functions of the affine parameter. The state vector is

$$Y(\lambda) = \begin{pmatrix} r(\lambda) \\ \phi(\lambda) \end{pmatrix}. \quad (57)$$

This state vector contains only the two coordinates needed to draw the photon path in the equatorial plane. For a fixed impact parameter b , the right-hand side of the system is

$$\frac{dY}{d\lambda} = \begin{pmatrix} s_r \sqrt{\mathcal{R}(r; b)} \\ b/r^2 \end{pmatrix}. \quad (58)$$

The sign s_r is not optional. Since the radial equation gives $(dr/d\lambda)^2$, the code must specify whether the photon is moving inward or outward.

The integration evolves r and ϕ , rather than solving directly for $r(\phi)$. This choice matters because escaping trajectories change radial direction at a turning point. Before the turning point, $s_r = -1$. After the turning point, $s_r = +1$. Treating the two branches explicitly makes the sign change part of the algorithm, rather than leaving it to the numerical solver.

All calculations use geometric units with

$$G = c = M = 1.$$

Distances are therefore measured in units of the black-hole mass. With this convention, the Schwarzschild radius is $2M = 2$, and the analytic critical impact parameter is $b_{\text{crit}} = 3\sqrt{3}M \approx 5.196152M$.

The initial position is chosen as

$$r(0) = r_0, \quad \phi(0) = 0.$$

The value $\phi_0 = 0$ places the photon on the positive x -axis in the trajectory plots. It does not imply radial motion. The initial direction is set by

$$\left. \frac{dr}{d\lambda} \right|_{\lambda=0} = -\sqrt{\mathcal{R}(r_0; b)}, \quad \left. \frac{d\phi}{d\lambda} \right|_{\lambda=0} = \frac{b}{r_0^2}. \quad (59)$$

The first expression makes each photon start on the inward branch. The second expression shows how the impact parameter sets the initial angular motion. Only $b = 0$ gives a purely radial photon path.

For the trajectory plots, I used $r_0 = 30M$. This places the launch point outside the strong-field region while keeping the plotted trajectories compact. For the deflection-angle calculation, I used $r_0 = 100M$. The larger starting radius gives a better approximation to scattering from large distance, so it is more appropriate for comparison with weak-field deflection formulas.

The code classifies a ray as captured if it reaches

$$r = 2M + \varepsilon_h, \quad (60)$$

where $\varepsilon_h = 10^{-5}$. The small offset avoids stopping exactly at the Schwarzschild coordinate singularity. A ray is classified as escaping if it reaches the escape radius R_{esc} while on the outward branch.

The differential equations are integrated using the `solve_ivp` routine from SciPy [12]. Since no method is specified in the code, SciPy uses its default adaptive RK45 Runge–Kutta method. The solver chooses its internal step sizes subject to the relative and absolute tolerances, and I impose a maximum allowed step in λ . The main numerical settings are listed in Table 1.

Quantity	Value	Role in the calculation
Mass scale	$M = 1$	Sets geometric units
Initial radius for trajectory plots	$r_0 = 30M$	Starting radius for orbit visualisation
Initial radius for deflection plot	$r_0 = 100M$	Starting radius for scattering calculation
Initial angle	$\phi_0 = 0$	Places the ray on the positive x -axis
Initial radial branch	$s_r = -1$	Ray starts inward
Critical impact parameter	$b_{\text{crit}} = 3\sqrt{3}M$	Analytic capture–escape benchmark
Escape radius	$R_{\text{esc}} = 100M$	Ray is classified as escaping when reached outward
Horizon cut-off	$2M + \varepsilon_h$, with $\varepsilon_h = 10^{-5}$	Ray is classified as captured when reached inward
Relative tolerance	10^{-10}	Adaptive solver tolerance
Absolute tolerance	10^{-12}	Adaptive solver tolerance
Maximum step in λ	5×10^{-3}	Largest allowed affine-parameter step
Restart tolerance	10^{-4}	Distance outside the turning radius used for branch switching
Radicand tolerance	10^{-6}	Handles small negative round-off errors near turning points

Table 1: Numerical settings used in the ray-tracing calculation. The photon sphere is not an input to the integration algorithm; it is used later as a theoretical reference when interpreting near-critical trajectories.

3.1.2 Branch switching and tolerance choice

The square root in Eq. (58) makes the turning point numerically delicate. Radial motion is real only where

$$\mathcal{R}(r; b) \geq 0.$$

Near a turning point, $\mathcal{R}(r; b)$ approaches zero. In exact arithmetic this gives $dr/d\lambda = 0$. In floating-point arithmetic, however, round-off can make the radicand slightly negative even when the analytical trajectory is valid. In the code, small negative values with magnitude below $\text{TURN_TOL} = 10^{-6}$ are set to zero. Larger negative values are treated as unphysical.

The integration is performed branch by branch. For $b < b_{\text{crit}}$, no outer turning point exists for an incoming ray from large radius, so the photon is integrated inward with $s_r = -1$ until it reaches the horizon cut-off. For $b > b_{\text{crit}}$, the photon first moves inward, reaches a radial turning point, and then moves outward again. The exact case $b = b_{\text{crit}}$ is the limiting trajectory associated with the unstable photon sphere; in the finite numerical classification, it is treated with the captured branch rather than followed as an exact asymptotic orbit.

The turning radius is computed from the cubic equation derived in Eq. (40),

$$r^3 - b^2 r + 2Mb^2 = 0. \quad (61)$$

The code selects the largest real root outside the event horizon. This root gives the outer turning radius r_{turn} for an escaping trajectory.

The integration does not restart exactly at r_{turn} . At the exact turning point, the radicand vanishes, and the square root becomes sensitive to round-off. Instead, the inward branch stops at

$$r_{\text{stop}} = r_{\text{turn}} + \text{RESTART_TOL}. \quad (62)$$

The outward branch then begins from r_{stop} with $s_r = +1$. This procedure makes the radial sign change explicit and avoids forcing the solver to step through the square-root behaviour at $\mathcal{R} = 0$.

The full ray-tracing procedure is summarised in the following pseudocode.

Inputs:

```
M, r0, phi0, b_values, lambda_max
epsilon_horizon, R_escape
rtol, atol, max_step
RESTART_TOL, TURN_TOL
```

For each impact parameter b:

```
Set initial conditions:
```

```
  r = r0
  phi = phi0 = 0
```

```
Compute:
```

```
  bcrit = 3 sqrt(3) M
  R(r;b) = 1 - (1 - 2M/r) b^2/r^2
```

```
If b <= bcrit:
```

```
  Set radial branch:
```

```
    s_r = -1
```

```
  Integrate inward:
```

```

    dr/dlambda = -sqrt(R(r;b))
    dphi/dlambda = b/r^2

Stop if:
    r = 2M + epsilon_horizon

If the horizon cut-off is reached:
    classify ray as CAPTURE
Otherwise:
    classify ray as UNDECIDED

If b > bcrit:

    Compute r_turn from:
        r^3 - b^2 r + 2 M b^2 = 0

    Select the largest real root with:
        r_turn > 2M

    Set:
        r_stop = r_turn + RESTART_TOL

    Integrate inward to r_stop:
        s_r = -1
        dr/dlambda = -sqrt(R(r;b))
        dphi/dlambda = b/r^2

    Restart outward from r_stop:
        s_r = +1
        dr/dlambda = +sqrt(R(r;b))
        dphi/dlambda = b/r^2

    Stop if:
        r = R_escape

    If the escape radius is reached:
        classify ray as ESCAPE
    Otherwise:
        classify ray as UNDECIDED

During integration:

    If R(r;b) is slightly negative and |R(r;b)| < TURN_TOL:
        set R(r;b) = 0

    If R(r;b) is significantly negative:
        stop the integration as unphysical

Store:
    b
    lambda values

```

```

r(lambda)
phi(lambda)
outcome
turning radius, if present
total angular displacement

```

The pseudocode shows the main numerical choice in the ray tracer. Captured rays are integrated as one inward branch, while escaping rays are split into inward and outward branches. The split is not a change in the physical trajectory. It is a numerical way of handling the sign change in $dr/d\lambda$ at the turning point.

This branch-based method is especially important near b_{crit} . Near-critical rays can remain close to the photon sphere for a long interval, so small numerical differences near the turning point can accumulate into visible differences in angular displacement. Splitting the trajectory into inward and outward branches gives direct control over the sign change in $dr/d\lambda$.

To test the effect of the restart tolerance, I repeated a near-critical escaping trajectory with

$$b = 5.1962.$$

This value lies slightly above $b_{\text{crit}} = 3\sqrt{3}M \approx 5.19615M$. The final calculations use `RESTART_TOL` = 10^{-4} , and Table 2 shows how the extracted trajectory quantities change when this value is varied.

RESTART_TOL	Classification	r_{min}/M	$\Delta\phi$	$\delta\phi_{\text{num}}$
10^{-3}	escape	3.008436	13.090647	10.175125
10^{-4}	escape	3.007536	13.788830	10.873307
10^{-5}	escape	3.007446	14.012743	11.097221
10^{-6}	escape	3.007437	14.083653	11.168130

Table 2: Sensitivity of a near-critical escaping trajectory with $b = 5.1962$ to the restart tolerance. The restart tolerance controls how close to the analytical turning radius the inward integration stops before the outward branch is started.

The classification is stable across the tested restart tolerances. In every case, the ray is classified as escaping. The angular quantities are more sensitive, because they depend on how close the numerical trajectory is allowed to approach the turning point before the branch switch. This sensitivity is expected for near-critical rays: the trajectory spends a long angular interval near the photon sphere, so small changes near closest approach affect the accumulated angle.

The value `RESTART_TOL` = 10^{-4} is used in the final calculations. This choice gives stable capture–escape classification while avoiding numerical problems from evaluating the square root too close to zero. The tolerance test shows that the trajectory outcome is robust for the selected near-critical ray. It also shows that angular quantities near b_{crit} should be interpreted as numerical diagnostics rather than exact asymptotic observables.

I also tested the adaptive solver tolerances. Changing the relative tolerance from 10^{-8} to 10^{-12} and the absolute tolerance from 10^{-10} to 10^{-14} did not change the trajectory classification for the tested ray. The branch-switch tolerance therefore has the more visible effect on the near-critical angular quantities than the standard `solve_ivp` tolerances.

With the integration procedure fixed, the next step is to test the physical classification produced by the code. The central question is whether the numerical trajectories separate at the analytic value b_{crit} , as predicted by the effective-potential analysis.

3.2 Capture–Escape Classification of Photon Trajectories

The first numerical result tests whether the ray tracer reproduces the expected capture–escape split. For each selected impact parameter b , the code integrates the reduced equations for $r(\lambda)$ and $\phi(\lambda)$. A ray is classified as captured if it reaches the horizon cut-off $r = 2M + \varepsilon_h$. It is classified as escaping if it reaches the escape radius R_{esc} on the outward branch.

The analytic value

$$b_{\text{crit}} = 3\sqrt{3}M \approx 5.196152M$$

is used as the benchmark. The calculation does not determine this value independently. Instead, it checks whether the numerical trajectories behave as expected on either side of the theoretical boundary. Rays with $b < b_{\text{crit}}$ should be captured, while rays with $b > b_{\text{crit}}$ should reach a turning point and escape.

For plotting, the polar trajectory is converted to Cartesian coordinates by

$$x = r \cos \phi, \quad y = r \sin \phi. \quad (63)$$

These coordinates are used only to visualise the spatial path in the equatorial plane. They are not the observer-screen coordinates used later for the shadow construction.

Figure 1 shows the capture–escape structure across a broad range of impact parameters. The full plotted range covers $0 \leq b \leq 10$, while the labelled subset highlights values close to the critical impact parameter. This allows the global behaviour and the transition region to be shown in the same figure.

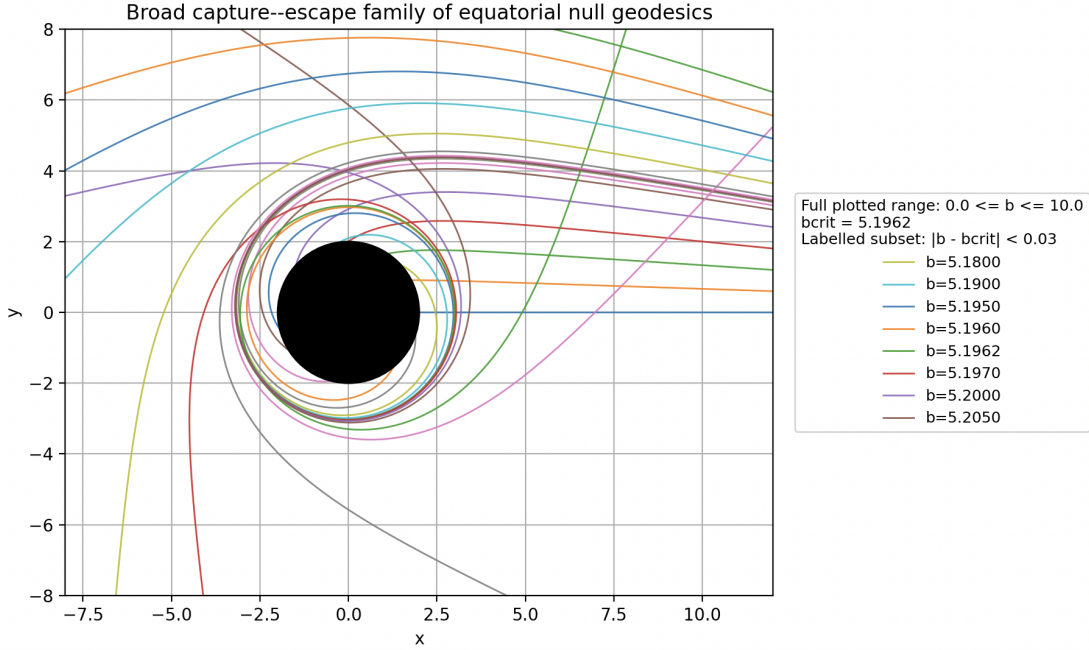


Figure 1: Broad capture–escape family of equatorial null geodesics in Schwarzschild spacetime. The full plotted range covers $0 \leq b \leq 10$, while the labelled trajectories show values near $b_{\text{crit}} \approx 5.1962M$. The black disk represents the event horizon at $r = 2M$. Rays with smaller impact parameter cross the horizon cut-off and are classified as captured. Rays with larger impact parameter reach a radial turning point and escape.

The trajectories separate into two groups. For smaller b , the photon has insufficient angular momentum relative to its energy to avoid the black hole. These rays continue inward until they reach the horizon cut-off. For larger b , the photon reaches a closest approach and then moves outward again. This agrees with the effective-potential result from Section 2: the critical impact parameter separates trajectories with no outer turning point from those that turn around and escape.

The labelled curves in Figure 1 also show why the near-critical region needs separate attention. Several trajectories pass close to the black hole and bend through a large angle before their final outcome becomes clear. The capture–escape classification is still binary, but the spatial paths near b_{crit} are not simple inward or outward curves. They can linger near the photon sphere before either crossing the horizon cut-off or escaping.

Table 3 gives representative classifications. The values $b = 5.1960$ and $b = 5.1962$ lie on opposite sides of b_{crit} . The first is slightly below the critical value and is captured. The second is slightly above it and escapes. This small change in b is enough to change the final classification, which is the numerical signature of the capture–escape boundary.

Impact parameter	Relation to b_{crit}	Classification
$b = 4.0000$	Below b_{crit}	Capture
$b = 5.1960$	Slightly below b_{crit}	Capture
$b = 5.1962$	Slightly above b_{crit}	Escape
$b = 6.0000$	Above b_{crit}	Escape

Table 3: Representative capture–escape classifications for selected impact parameters. The critical value is $b_{\text{crit}} = 3\sqrt{3}M \approx 5.196152M$.

This result answers the first part of the research question. The numerical ray tracer reproduces the expected transition between captured and escaping Schwarzschild null geodesics. The transition is controlled by the impact parameter b , with the boundary located at the analytically known value b_{crit} .

The classification alone, however, does not show why the transition is physically interesting. The near-critical cases in Table 3 differ only slightly in b , yet their final outcomes are different. The next subsection therefore examines the paths near this boundary in more detail.

3.3 Near-Critical Trajectories and Orbital Winding

The previous subsection showed where the capture–escape split occurs. I now focus on what the rays do near that split. This is the sensitive part of the problem, because small changes in b can change the final outcome while the early part of the trajectory remains visually similar. That behaviour is the numerical signature of the unstable photon sphere.

The near-critical range was sampled using impact parameters around

$$b_{\text{crit}} = 3\sqrt{3}M \approx 5.196152M.$$

Values slightly below this threshold are expected to be captured, while values slightly above it are expected to escape after reaching a turning point. The point of this calculation is not to redetermine b_{crit} , but to show how the trajectories behave when b is close to the analytically known capture–escape boundary.

Figure 2 shows a family of near-critical trajectories. The plotted rays pass close to the black hole before separating into captured and escaping branches. Compared with the broad family in Figure 1, these trajectories remain near the central region for longer and accumulate a larger angular displacement.

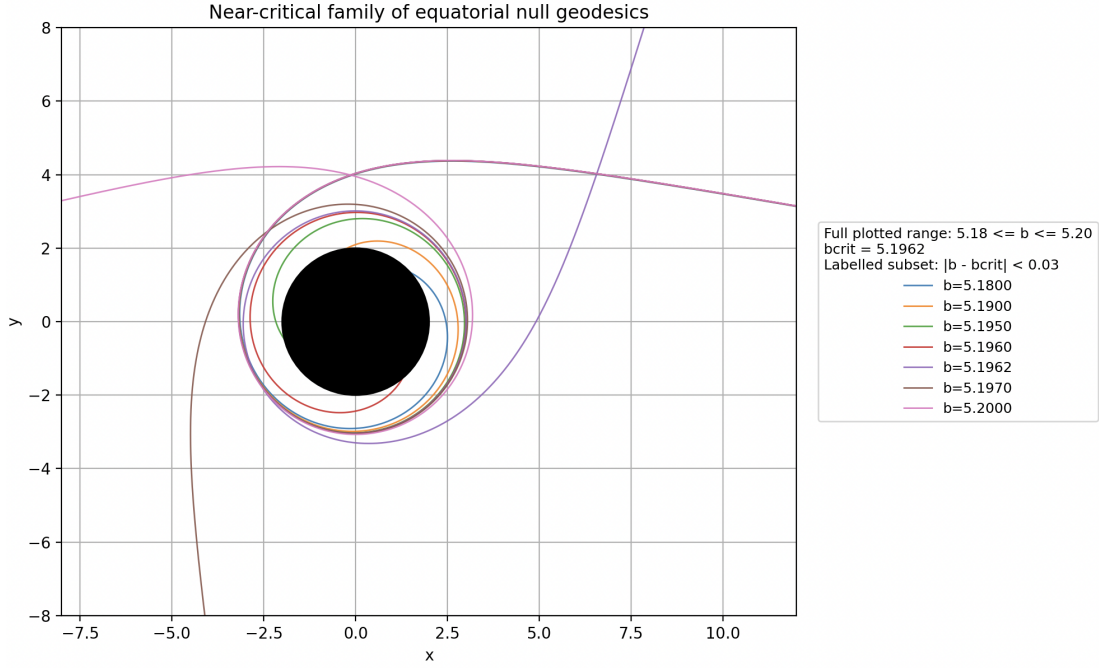


Figure 2: Near-critical family of equatorial null geodesics in Schwarzschild spacetime. The plotted values lie close to $b_{\text{crit}} \approx 5.1962M$. The black disk marks the event horizon at $r = 2M$. Rays near the critical impact parameter bend strongly near the black hole before separating into captured and escaping outcomes.

The clustering of paths near the black hole is the important feature of Figure 2. Away from b_{crit} , a ray either falls in more directly or escapes with a smaller angular change. Near the critical value, the ray passes close to the unstable photon sphere, so the angular coordinate continues to evolve before the final branch is decided. This behaviour is the numerical form of the near-critical lensing discussed in Section 2.

To make the transition clearer, Figure 3 compares two representative trajectories using the same axis limits. The first has

$$b = 5.1960 < b_{\text{crit}},$$

and is captured. The second has

$$b = 5.1962 > b_{\text{crit}},$$

and escapes. The two values differ only in the fourth decimal place, yet they end on opposite sides of the capture–escape boundary.

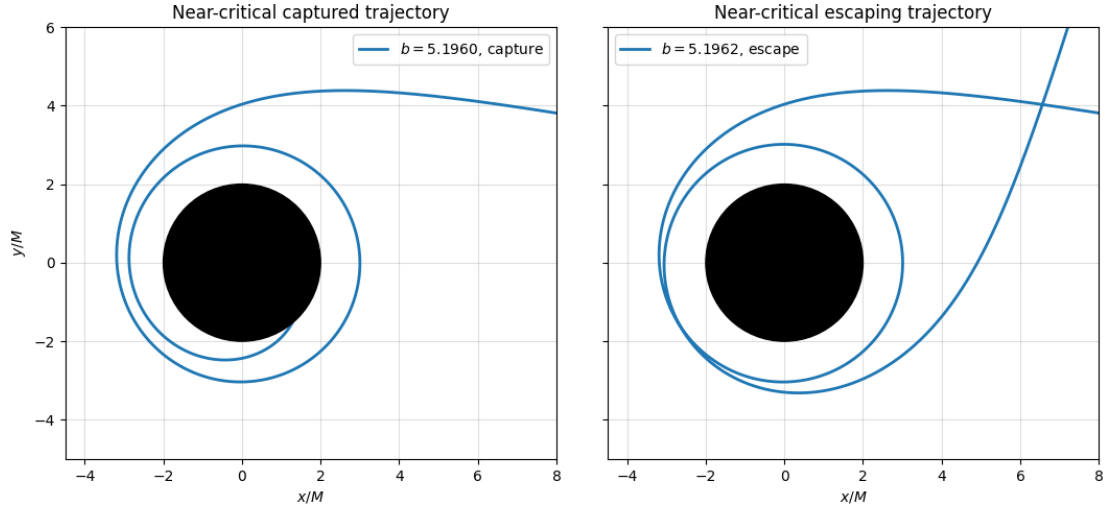


Figure 3: Two representative near-critical photon trajectories shown with the same axis limits. The black disk marks the event horizon at $r = 2M$. Left: $b = 5.1960 < b_{\text{crit}}$, which is captured. Right: $b = 5.1962 > b_{\text{crit}}$, which escapes after passing through the near-black-hole region. The common axis limits make the near-critical behaviour directly comparable.

The two trajectories initially follow similar paths because both pass close to the unstable photon sphere. Their final outcomes differ only after this near-critical part of the motion. The captured ray eventually crosses the horizon cut-off, while the escaping ray reaches a turning point and moves outward. This comparison shows why the capture–escape boundary is numerically delicate: close to b_{crit} , the trajectory can spend a long interval near the same unstable region before the final outcome becomes clear.

To quantify the amount of angular motion, I compute the total angular displacement of each stored trajectory,

$$\Delta\phi = |\phi_{\text{final}} - \phi_{\text{initial}}|. \quad (64)$$

From this, I define an approximate winding number,

$$N \approx \frac{\Delta\phi}{2\pi}. \quad (65)$$

This is not an exact count of circular orbits at the photon sphere. It includes the angular change during approach and, for escaping rays, during the outward branch. I use it as a numerical diagnostic for how much the photon winds around the black hole.

Figure 4 shows N as a function of impact parameter near b_{crit} . The dashed vertical line marks the analytical critical value.

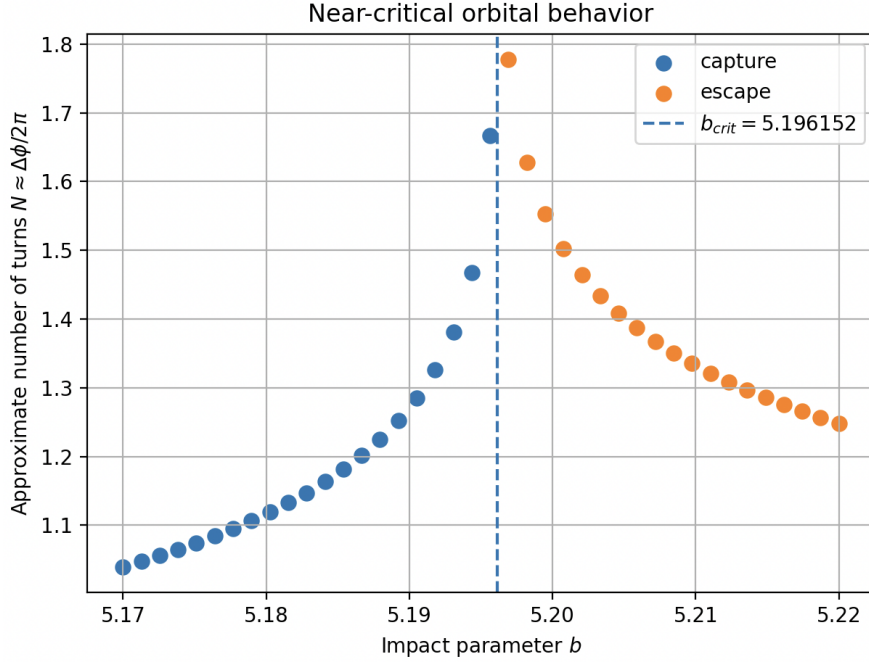


Figure 4: Approximate winding number $N \approx \Delta\phi/(2\pi)$ as a function of impact parameter near b_{crit} . The dashed vertical line marks the critical impact parameter. The increase near this value shows that near-critical rays accumulate a larger angular displacement before capture or escape.

The winding number increases near b_{crit} . This does not mean that the numerical ray reaches a stable circular orbit. No stable circular photon orbit exists in Schwarzschild spacetime. Instead, the increase in N shows that the ray spends longer near the unstable photon sphere before leaving the near-critical region. For $b < b_{\text{crit}}$, the photon eventually crosses the horizon cut-off. For $b > b_{\text{crit}}$, it turns around and escapes.

This result answers the near-critical part of the research question. The capture–escape transition is not only a binary classification of rays. Close to b_{crit} , both captured and escaping trajectories can undergo large angular bending before their final outcomes differ. This is the same strong-field behaviour that produces the rapid growth of the deflection angle near the capture threshold [7, 8, 9].

The next diagnostic tests the same near-critical behaviour in a different way. Instead of counting angular winding for both captured and escaping rays, I compute the deflection angle only for rays that escape back to large radius.

3.4 Deflection Angle

The deflection angle gives a scattering-based view of the same physics. It is defined only for escaping photon trajectories, because captured photons do not return to large radius and therefore do not have an outgoing asymptotic branch. This makes the deflection plot complementary to the winding-number plot: it focuses on the escaping side of the capture–escape boundary.

For each escaping ray, the code records the total angular change along the numerical trajectory,

$$\Delta\phi_{\text{GR}} = |\phi_{\text{final}} - \phi_{\text{initial}}|. \quad (66)$$

In the ideal scattering problem, the photon starts at infinity, reaches a closest approach, and escapes back to infinity. In that limit, the flat-space reference angular change is π , and the deflection angle is $\delta\phi = \Delta\phi - \pi$, as in Eq. (49).

The numerical calculation uses finite radii instead. The photon starts at $r_{\text{start}} = 100M$ and is stopped when it reaches $r_{\text{end}} = 100M$ on the outward branch. Subtracting exactly π would therefore mix gravitational deflection with a finite-distance geometric effect. To remove this, I subtract the angular change of a straight-line path over the same finite radial interval. For a flat-space trajectory with impact parameter b , this reference angle is

$$\Delta\phi_{\text{flat}} = \arccos\left(\frac{b}{r_{\text{start}}}\right) + \arccos\left(\frac{b}{r_{\text{end}}}\right). \quad (67)$$

The finite-radius corrected numerical deflection angle is then

$$\delta\phi_{\text{num}} = \Delta\phi_{\text{GR}} - \Delta\phi_{\text{flat}}. \quad (68)$$

This definition compares the curved-spacetime trajectory with a flat-spacetime trajectory using the same start and end radii. It is therefore better matched to the actual numerical setup than the asymptotic subtraction by π .

The numerical values are compared with the weak-field expressions derived in Section 2. For large impact parameter, the leading-order Schwarzschild deflection is

$$\delta\phi \approx \frac{4M}{b},$$

while the second-order weak-field approximation includes the correction

$$\delta\phi \approx \frac{4M}{b} + \frac{15\pi M^2}{4b^2}.$$

These approximations are expected to work when $b \gg M$, where the ray remains far from the black hole [1, 2, 11]. They should fail near b_{crit} , where the photon enters the strong-field region.

Figure 5 shows the finite-radius corrected deflection angle for escaping trajectories. The numerical points are produced by the Python ray tracer. The dashed and dotted curves show the leading-order and second-order weak-field approximations. The vertical dashed line marks the critical impact parameter.

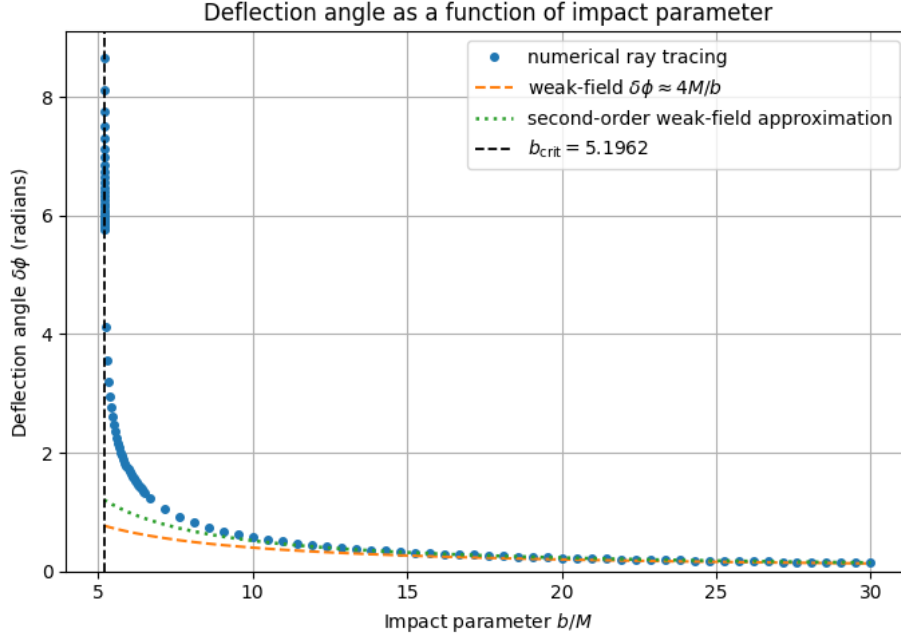


Figure 5: Finite-radius corrected deflection angle for escaping photon trajectories as a function of impact parameter. The numerical points are produced by the ray-tracing code. The dashed curve shows the leading-order weak-field approximation $\delta\phi \approx 4M/b$, while the dotted curve shows the second-order weak-field approximation. The vertical dashed line marks b_{crit}/M .

For large b , the numerical points follow the weak-field curves closely. This is the expected behaviour: the photon passes far from the black hole, the curvature is weak, and the bending angle is small. The second-order expression gives a better comparison than the leading-order expression because it keeps the next term in the expansion in M/b .

As b decreases toward b_{crit} , the numerical deflection angle rises rapidly. In this range, the photon passes close to the photon sphere before escaping. The weak-field formulas no longer describe the trajectory because their expansion assumes $M/b \ll 1$, while near-critical rays are controlled by the strong-field geometry close to $r = 3M$.

This result is consistent with the near-critical trajectories in Section 3.3. The same mechanism appears in both diagnostics. Near b_{crit} , the ray spends a longer angular interval near the unstable photon sphere, so both the winding number and the deflection angle increase. This is also the behaviour described in strong-field lensing treatments, where the deflection angle grows rapidly as the capture threshold is approached [7, 8].

The deflection-angle plot therefore validates the ray tracer in two regimes. In the weak-field regime, the numerical output agrees with analytic approximations. In the strong-field regime, the numerical output departs from those approximations in the expected direction and shows the rapid growth associated with near-critical photon motion.

The same capture–escape structure now has to be translated from coordinate-space trajectories into an

observer-screen image. That translation is the role of the shadow construction.

3.5 The Schwarzschild Shadow

The shadow calculation is not separate from the previous trajectory results. It is their screen-space consequence. Sections 3.2–3.4 showed that varying b separates null geodesics into captured and escaping families, with near-critical rays controlled by the unstable photon sphere. The shadow is obtained by mapping that same classification onto an observer’s image screen.

The key change is the space in which the result is represented. The previous plots show photon paths in Schwarzschild coordinate space. The shadow is defined in impact-parameter space. A point on the observer’s screen belongs to the shadow if the corresponding backward-traced null geodesic is captured by the black hole. Thus, the shadow is the visual form of the capture–escape classification.

3.5.1 Shadow construction in impact-parameter space

The observer’s screen is described using two impact-parameter coordinates, b_x and b_y . The distance from the centre of the screen is

$$B = \sqrt{b_x^2 + b_y^2}. \quad (69)$$

This B is not a Schwarzschild radial coordinate. It is a radius in screen space, measured in units of impact parameter.

For a Schwarzschild black hole, spherical symmetry makes the capture condition depend only on B , not on the angular direction around the screen. The idealised shadow is therefore constructed using

$$B < b_{\text{crit}}. \quad (70)$$

Pixels satisfying this condition are assigned to the captured region. Pixels with $B > b_{\text{crit}}$ correspond to escaping directions. The resulting shadow boundary is

$$B = b_{\text{crit}} = 3\sqrt{3}M. \quad (71)$$

This construction does not require tracing every screen pixel separately, because the Schwarzschild case is circular by symmetry. The individual trajectory calculations still matter, however. They show why the screen boundary has this interpretation: $B = b_{\text{crit}}$ is the image-screen version of the capture–escape threshold found from the null geodesics.

3.5.2 Shadow formation from photon trajectories

Figure 6 shows a coordinate-space visualisation of the photon trajectories that produce the shadow boundary. This figure is not the observer’s screen. It shows the ray behaviour in the equatorial plane around the black hole.

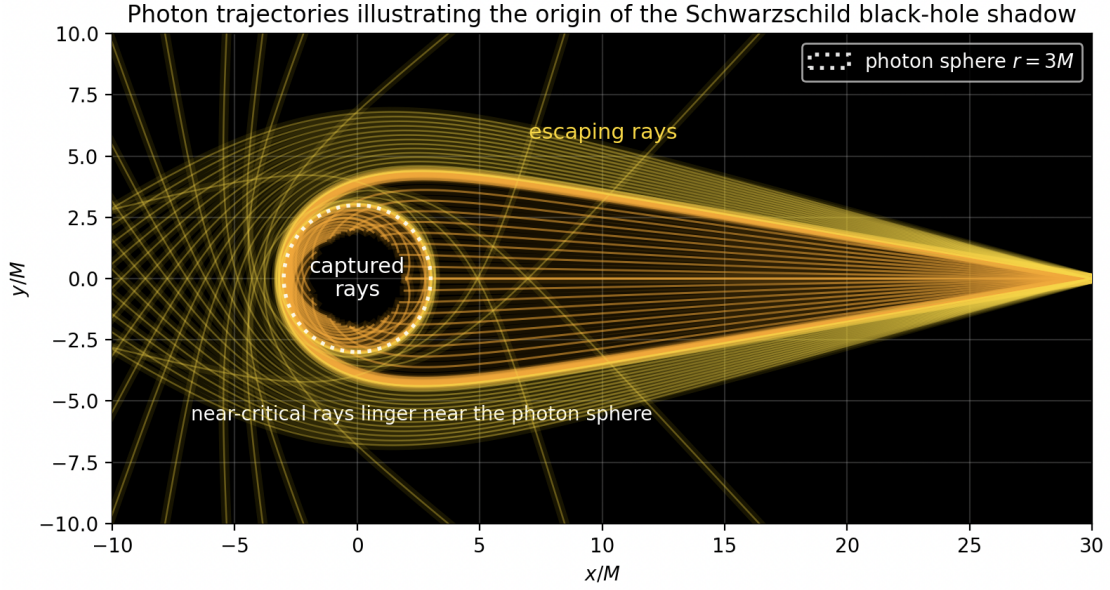


Figure 6: Coordinate-space visualisation of photon trajectories illustrating the origin of the Schwarzschild shadow. The black central region represents captured rays, and the dotted circle marks the photon sphere at $r = 3M$. Escaping rays leave the strong-field region, while near-critical rays linger close to the photon sphere before either escaping or being captured. The figure explains the geodesic behaviour behind the shadow boundary; it is not itself an observer-screen image.

The figure links the trajectory results to the screen-space shadow. Captured rays form the dark set of directions because no external light reaches the observer along those paths. Escaping rays can be traced back to external regions and therefore lie outside the shadow. Near-critical rays sit between these behaviours. They pass close to the photon sphere and mark the transition between captured and escaping directions.

This is the geometric origin of the Schwarzschild shadow boundary. The event horizon is drawn in coordinate space, but the shadow radius is not the horizon radius. The boundary on the observer's screen is set by $B = b_{\text{crit}}$, which comes from the limiting null geodesics associated with the unstable photon sphere.

3.5.3 Angular size of the shadow

The same shadow boundary can be expressed as an angular size for a static observer at radius r_{obs} . Using Eq. (55), the angular radius satisfies

$$\sin \alpha_{\text{sh}} = \frac{b_{\text{crit}}}{r_{\text{obs}}} \sqrt{1 - \frac{2M}{r_{\text{obs}}}}. \quad (72)$$

I evaluated this expression for static observers outside the photon sphere.

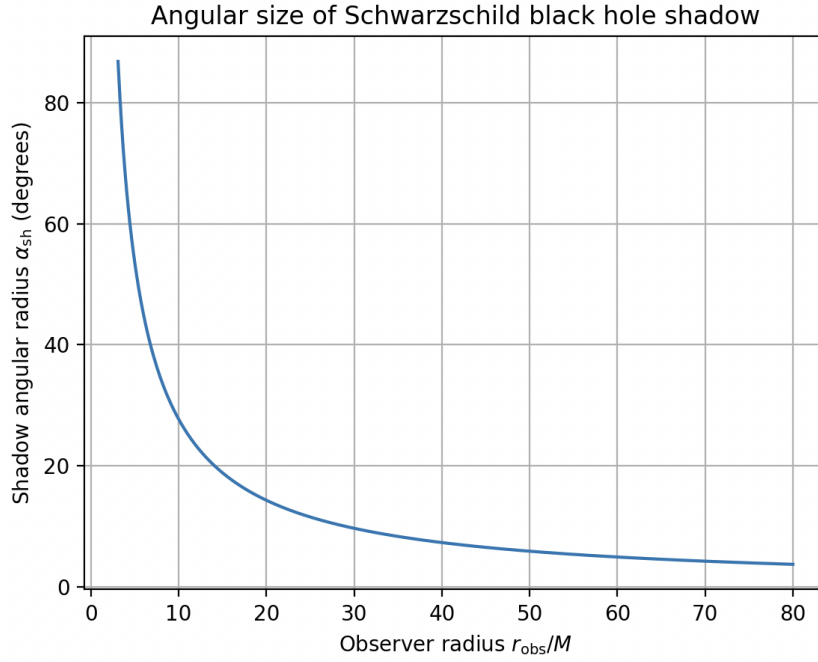


Figure 7: Angular radius of the Schwarzschild black-hole shadow as a function of observer radius for static observers outside the photon sphere. The apparent angular size decreases as the observer moves farther away from the black hole.

Figure 7 shows that the angular radius decreases with observer distance. This behaviour follows directly from Eq. (72). A nearby static observer sees the captured region cover a large part of the sky. A distant observer sees the same impact-parameter boundary under a smaller angle.

At large r_{obs} , the result approaches the small-angle relation

$$\alpha_{\text{sh}} \approx \frac{b_{\text{crit}}}{r_{\text{obs}}}.$$

This limit connects the local angular size to the screen-space radius $B = b_{\text{crit}}$. The plotted curve therefore shows the same shadow boundary in a different form: not as a radius on an image screen, but as an angle on the observer's sky.

3.5.4 Illustrative EHT-style image

As a final visualisation, I generated a simplified EHT-style image to connect the idealised Schwarzschild shadow to the broader visual language of black-hole imaging. The purpose of this figure is interpretative. It shows how a dark central region and a surrounding ring can be placed around the geometric shadow scale.

The image is built from a screen-based brightness prescription. A dark central region is placed inside the shadow scale, a bright ring is placed near the critical impact parameter, azimuthal brightness variations are added by hand, and a Gaussian blur is applied to imitate finite resolution. These choices turn the

geometric shadow scale into a qualitative image, while the quantitative interpretation remains tied to the geodesic calculations above.

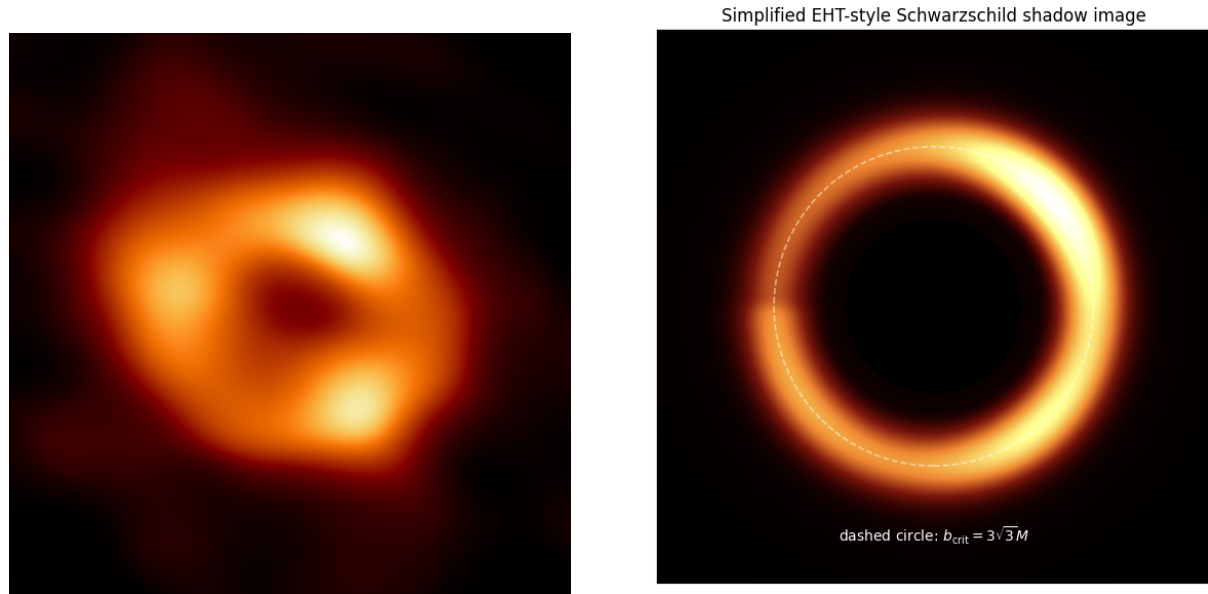


Figure 8: Qualitative comparison between the Event Horizon Telescope image of Sagittarius A* (left) and the illustrative Schwarzschild shadow image generated in this thesis (right). The right-hand image is based on a simple brightness prescription: a dark central region, a bright ring near the critical impact parameter, manually added azimuthal brightness variations, and Gaussian blurring. Image credit for Sagittarius A*: Event Horizon Telescope Collaboration [4].

Figure 8 should be read as a visual comparison, not as a validation plot. The EHT image comes from interferometric observations and depends on emission physics, relativistic plasma dynamics, and image reconstruction. The image generated here instead illustrates the geometric point established by the ray tracing: the central dark region is associated with captured photon directions, while the bright ring is placed near the critical shadow scale.

The quantitative results of this thesis remain the photon trajectories, the capture–escape classification, the near-critical winding, the deflection-angle calculation, and the shadow boundary $B = b_{\text{crit}}$. The EHT-style image is therefore a visual bridge between the idealised Schwarzschild calculation and the observational images that motivated the project.

Together, these shadow calculations answer the final part of the research question. The idealised Schwarzschild shadow is the screen-space representation of the captured null-geodesic region. Its boundary is fixed by the critical impact parameter, not by the event horizon radius. The numerical trajectories explain why this happens: rays near the boundary are the near-critical geodesics associated with the unstable photon sphere.

4 Discussion

Section 3 showed the numerical outputs of the ray tracer: photon paths, capture–escape classifications, winding estimates, deflection angles, and the screen-space shadow. I now interpret what those outputs mean physically, and where their limits are. The discussion therefore focuses on interpretation rather than on re-describing each figure.

The main result is not that the Schwarzschild shadow is circular. That follows from spherical symmetry. The main result is that the ray tracer reproduced the capture–escape transition that gives the shadow boundary its physical meaning. In the simulation, the shadow edge was not drawn by the event horizon. It appeared because photon trajectories on opposite sides of b_{crit} had different final outcomes.

This separates three related objects. The event horizon at $r = 2M$ is a causal boundary. The photon sphere at $r = 3M$ is an unstable null orbit. The shadow is a region on an observer’s screen. The ray-tracing calculation links these objects by following null geodesics and asking whether each one escapes or is captured. This is the interpretation used in the rest of the discussion.

4.1 Interpretation of the Capture–Escape Transition

The capture–escape transition is the physical core of the calculation. In Schwarzschild spacetime, the impact parameter

$$b = \frac{L}{E}$$

sets the angular momentum of a photon relative to its energy. Changing b changes whether the photon can reach an outer radial turning point. The analytic threshold is

$$b_{\text{crit}} = 3\sqrt{3}M,$$

which follows from the effective-potential analysis of Schwarzschild null geodesics [7, 6].

The numerical results support this picture. Rays with $b < b_{\text{crit}}$ did not reach an outer turning point. They moved inward until the integration stopped at the horizon cut-off. Rays with $b > b_{\text{crit}}$ reached a closest approach, switched radial branch, and escaped outward. The code therefore reproduced the expected split in photon outcomes.

This result should be read as a validation of the ray tracer, not as a new measurement of b_{crit} . I used the analytic value as an input benchmark. The useful result is that the reduced implementation behaved correctly on both sides of that benchmark. This matters because the later shadow construction assumes that b_{crit} can be treated as a boundary in photon initial conditions.

The trajectory plots make the analytic result easier to interpret. The effective potential tells us where the threshold lies. The ray tracer shows what the threshold looks like as a family of paths. Far below b_{crit} , rays fall in directly. Far above it, they bend and escape. Close to b_{crit} , the outcome becomes delayed because the ray spends longer near the strong-field region.

The horizon cut-off also has a clear numerical meaning. I stopped captured rays at

$$r = 2M + \varepsilon_h,$$

rather than at $r = 2M$, to avoid the Schwarzschild coordinate singularity. For the exterior problem, this does not change the classification. Once an inward-moving ray reaches this region, its capture outcome has already been decided.

4.2 Near-Critical Behaviour and the Photon Sphere

The most informative rays are the near-critical ones. Far from the threshold, the classification is visually straightforward. Near b_{crit} , small changes in b can produce different final outcomes after very similar early motion.

The pair $b = 5.1960$ and $b = 5.1962$ shows this clearly. These values lie on opposite sides of

$$b_{\text{crit}} \approx 5.196152M.$$

Their trajectories initially pass through the same near-black-hole region, but one is captured and the other escapes. A small change in impact parameter moves the photon from one side of the separatrix to the other.

This sensitivity comes from the unstable photon sphere at $r = 3M$. The exact critical ray is a limiting trajectory associated with this radius. Nearby rays do not settle into a stable circular orbit. They remain near the photon sphere for some angular interval and then leave. One side falls inward. The other side turns around and escapes [9, 6].

The winding estimate measures this delayed departure. I used

$$N \approx \frac{\Delta\phi}{2\pi}$$

as a diagnostic for accumulated angular motion. It is not a literal count of stable orbits. A larger value of N only means that the photon accumulated more angular displacement before its outcome was reached. The increase near b_{crit} therefore indicates near-critical motion around the unstable photon sphere.

The deflection-angle calculation gives the same interpretation from the escaping side. For large b , the photon stays far from the black hole, so the numerical deflection follows the weak-field formulas. As b approaches b_{crit} from above, the photon passes closer to $r = 3M$, and the deflection grows rapidly. The weak-field approximation then fails because its assumption $M/b \ll 1$ no longer describes the path well.

This trend agrees with the strong-field lensing literature. Darwin studied strong bending in Schwarzschild spacetime, and Bozza showed that the deflection angle grows logarithmically near the capture threshold [7, 8]. My calculation does not resolve the exact asymptotic divergence. It uses finite launch and escape radii, finite solver tolerances, and a discrete set of impact parameters. It does show the expected trend: escaping rays bend much more strongly as they approach the capture boundary.

4.3 From Null Geodesics to the Schwarzschild Shadow

The shadow construction is the screen-space version of the same classification. A point on the observer's screen belongs to the idealised shadow if its corresponding null geodesic is captured. For Schwarzschild

spacetime, spherical symmetry makes the condition depend only on the screen radius

$$B = \sqrt{b_x^2 + b_y^2}.$$

The shadow edge is therefore

$$B = b_{\text{crit}}.$$

This gives the shadow radius a precise meaning. It is not the event horizon radius. The horizon radius $2M$ and photon-sphere radius $3M$ are Schwarzschild coordinate radii. The shadow radius $B = b_{\text{crit}}$ is an impact-parameter radius on the observer's screen. These quantities have different meanings, even though they are all measured in units of M .

This distinction matters when reading the figures. The black disk in the trajectory plots marks the horizon in coordinate space. The circular boundary in the shadow plot marks the capture threshold in screen space. The first is drawn in the orbital plane. The second is drawn in impact-parameter space. Confusing the two would make the shadow look like a projected image of the horizon, which it is not [10, 9, 6].

The angular-size calculation expresses the same boundary for a static observer at radius r_{obs} :

$$\sin \alpha_{\text{sh}} = \frac{b_{\text{crit}}}{r_{\text{obs}}} \sqrt{1 - \frac{2M}{r_{\text{obs}}}}.$$

As r_{obs} increases, the same impact-parameter scale covers a smaller angle on the observer's sky. At large radius, this reduces to

$$\alpha_{\text{sh}} \approx \frac{b_{\text{crit}}}{r_{\text{obs}}}.$$

The screen-space shadow and the angular shadow are therefore two descriptions of the same capture boundary [6].

The shadow plot is therefore not just a plotted circle. It translates the ray classifications into a set of dark directions. The numerical trajectories explain why the boundary of that set is controlled by near-critical geodesics rather than by the horizon surface.

4.4 Numerical Interpretation and Scope

The most robust output of the code is the binary classification of rays. Across the tested impact parameters, the ray tracer separated capture and escape according to the analytic threshold b_{crit} . This directly answers the computational part of the research question.

The angular quantities need more caution. Near b_{crit} , a ray can spend a long angular interval close to the photon sphere. Small numerical choices can then affect the accumulated angle. This appeared in the restart-tolerance test: the selected near-critical ray remained classified as escaping, but $\Delta\phi$, N , and $\delta\phi_{\text{num}}$ changed with the restart tolerance.

This sensitivity also explains why I treated the near-critical angular quantities differently from the capture–escape classification. The classification only asks which final state the ray reaches. The angular quantities depend on how long the ray remains near the photon sphere before that final state is reached. Since this time can grow rapidly near the separatrix, small numerical choices have a larger visible effect

on $\Delta\phi$ than on the final outcome. For that reason, I used the angular quantities to identify near-critical behaviour, not to extract a precise limiting value.

This does not weaken the main result. It shows that classification is more stable than the exact near-critical angle in a finite calculation. This is expected. The critical trajectory is a limiting case, so angular accumulation becomes increasingly sensitive near the separatrix. The winding and deflection plots should therefore be read as diagnostics of near-critical behaviour, not as exact asymptotic observables.

The use of Schwarzschild spacetime also limits the physical claim. The spacetime is static and spherically symmetric, so the idealised shadow is circular. This makes it a clean baseline for studying photon capture. It does not describe the full shadow of a rotating black hole. In Kerr spacetime, spin and observer inclination change the photon region and can alter the apparent boundary [9, 6].

The model also traces photon paths only. It does not solve radiative transfer. It therefore gives the geometric capture region, not a physical brightness map. Real intensities depend on emission, absorption, plasma effects, magnetic fields, and observer-dependent Doppler effects. Those ingredients are outside the scope of this thesis.

4.5 Connection to Observational Black-Hole Images

The EHT-style image in Section 3 should be read as an illustration. It uses the visual language of black-hole imaging, with a dark centre and surrounding emission, but the physical calculation behind it is still the Schwarzschild capture boundary. The brightness pattern was imposed by a simplified screen prescription.

Real Event Horizon Telescope images contain much more physics. The images of M87* and Sagittarius A* are reconstructed from interferometric data. They depend on plasma emission, relativistic beaming, optical depth, magnetic fields, variability, and image reconstruction choices [3, 4]. The observed ring is therefore not identical to the idealised mathematical shadow, although both depend on general-relativistic light propagation [3, 4, 6].

The comparison is still useful. Black-hole images are complicated, but their interpretation starts with geodesic structure. Before adding emission models or telescope reconstruction, one needs to know which photon directions escape and which are captured. This thesis isolates that geometric part of the problem in Schwarzschild spacetime.

Within that scope, the conclusion is clear. The Schwarzschild shadow is the observer-screen representation of captured null geodesics. Its boundary is fixed by the critical impact parameter, not by the event horizon radius. The near-critical trajectories provide the link between the geodesic equations and the dark region on the screen.

5 Conclusion and Further Directions

This thesis investigated how numerical ray tracing of Schwarzschild null geodesics reproduces the transition between photon capture and escape, and how that transition defines the boundary of an idealised black-hole shadow. The calculation used the analytic critical impact parameter,

$$b_{\text{crit}} = 3\sqrt{3}M,$$

as a benchmark for the ray tracer.

The simulation reproduced the expected capture–escape split. Photons with $b < b_{\text{crit}}$ reached the horizon cut-off and were classified as captured. Photons with $b > b_{\text{crit}}$ reached a radial turning point, switched to the outward radial branch, and escaped. The numerical results therefore confirmed the main theoretical picture: the shadow boundary is set by a threshold in photon trajectories, not by the event horizon radius itself.

The near-critical trajectories showed why this threshold is physically important. Rays close to b_{crit} passed near the unstable photon sphere at $r = 3M$ and accumulated large angular displacement before their final outcome was reached. A small change in impact parameter could move a photon from capture to escape. The winding estimates and deflection-angle calculations both reflected this same behaviour. They showed that near-critical rays are delayed near the photon sphere, while weak-field approximations break down as the capture threshold is approached.

The shadow construction translated this trajectory behaviour into observer-screen space. In the idealised Schwarzschild case, a screen point belongs to the shadow when the corresponding null geodesic is captured. Because of spherical symmetry, the boundary is circular and occurs at

$$B = b_{\text{crit}}.$$

This means that the idealised Schwarzschild shadow is not a direct image of the event horizon at $r = 2M$. It is the observer-screen representation of photon capture, controlled by near-critical null geodesics associated with the photon sphere.

The numerical method was useful because it made this analytic structure visible. The reduced conserved-quantity formulation allowed the impact parameter, radial turning points, and branch switching to appear directly in the code. This was especially useful near b_{crit} , where the radial radicand approaches zero and the integration becomes sensitive to finite numerical choices.

The main limitation is that the calculation was designed to study geometry, not full image formation. I used Schwarzschild spacetime, so the black hole was non-rotating and spherically symmetric. I also traced photon paths only, without modelling emission, absorption, plasma effects, or telescope reconstruction. The result is therefore an idealised shadow boundary, not a realistic Event Horizon Telescope image. The classification of capture and escape is the most robust numerical output. Quantities such as $\Delta\phi$, N , and $\delta\phi_{\text{num}}$ should be read as finite-radius diagnostics of near-critical behaviour rather than exact asymptotic observables.

Several extensions follow naturally from this work. One improvement would be to locate the capture–escape boundary numerically by scanning over b and applying a bisection method. This would test the separatrix directly instead of using the analytic Schwarzschild value of b_{crit} as an input. The same approach would be useful in spacetimes where the critical impact parameter is not available in closed form.

A second extension would be to replace the Schwarzschild metric with the Kerr metric. Spin breaks the spherical symmetry used in this thesis, changes the photon regions, and makes the apparent shadow depend on observer inclination [9, 6]. A Kerr ray tracer would therefore show how rotation changes the clean circular boundary found here.

A third direction would be to add a simple emission model, such as a thin accretion disc. That would move the project from photon geometry toward image formation. It would also connect the ray-tracing framework more directly to Luminet's black-hole visualisation and to modern Event Horizon Telescope images [5, 3, 4]. In such a model, the geodesics would still determine photon paths, but the observed brightness would also depend on where the light is emitted.

The central result is therefore simple: the idealised Schwarzschild shadow is the observer-screen image of photon capture. Its boundary is fixed by $b_{\text{crit}} = 3\sqrt{3}M$, not by the horizon radius. Numerical ray tracing makes this geometric mechanism visible by showing how null geodesics near the unstable photon sphere separate into captured and escaping paths.

A AI Statement

This thesis was prepared with limited assistance from artificial intelligence tools, mainly ChatGPT. The use of AI was within the context of the Senior Project AI policy, which permits AI use up to Level 3 unless otherwise specified.

AI was used as a thesis-support tool rather than as a replacement for the research process. In particular, I used AI to help formulate and refine the research question, to think through the overall structure of the thesis, and to decide which sections and subsections would be useful for presenting the project clearly.

In the later stages of writing, I used AI in a thesis-coaching role, similar to the approach encouraged in Dr Rensma's thesis coach prompt. This involved asking for feedback on whether the introduction and conclusion matched the aims of the thesis, whether the sections followed a logical order, and whether the overall structure was clear for a reader.

I treated AI output critically and used it only as guidance. A record of relevant prompts and outputs can be made available if requested by the exam board.

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